

Date : _____

No SQL

Database is a systematic collection of data & Database management system support storage and manipulation of data.

for eg.

Telephonic directory uses database to store data of people, their phone no. & other details that details can be used by service providers to manage billing, client related issues.

DBMS provides the mechanism to store & retrieve the data.

Different kinds of DBMS

→ RDBMS

→ OLAP (online analytical processing.)

→ NOSQL (Not only SQL)

Why & NO SQL

No SQL represent does not represent ^{single} ~~SAT~~ product or technology but it represent group of projects & various related data concepts for storage & management but they can accommodate a wide variety of data models. including key value document column & graph formats.

1998 ~~that~~ Karl Straussie introduced NoSQL

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NoSQL is ~~non~~ non relational

Alternative is to RDBMS, file set of database.
Provides flexible of schemas & for storage
& retrieval of data by beyond the traditional
table structures found in RDBMS.

It relates to large data sets accessed &
manipulated on a web.

NoSQL

Concept of NoSQL database become popular
with internet giants like google facebook
amazon etc who deal with huge volumes
system response very slow with RDBMS to
resolve this problem we could scale up our
system by upgrading our existing hw
this process is very expensive.

So alternative for this to distribute this
load on multiple hosts whenever load increases
this method is called scaling.

NoSQL database is designed for web application

NoSQL handle various types of data

- Semi structured
- Unstructured
- Rapidly changing data
- Big data

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features

No sql not after follow

- ① Relational model & table with flat fixed Column records
- ② They doesn't require object oriented mapping & data normalization
- ③ Most of the NoSQL database are Open Source
- ④ Most of NoSQL database executed in distributed fashion & they offer auto scaling
- ⑤ They follow shared nothing architecture, which require less coordination with higher distribution.
- ⑥ It offers heterogeneous structure of data in the same domain.
- ⑦ It is schemaless
- ⑧ No sql databases offers easy to use interfaces for storage & querying data provided

SQL	NO SQL
① Relational database management system	Non relational or distributed database system
② Have fixed or static or predefined schema	Have dynamic schema
③ Not suited for hierarchical data storage	Best suited for hierarchical data storage
④ Vertically Scalable	Horizontally Scalable
⑤ follows ACID Property atomicity, consistency, integrity, durability	Follow CAP (Consistency, availability, Partition tolerance)

CAP theorem

Also called as Brewer's theorem

^{NoSQL}
Different types of databases are available based on data models.

Data models.

Data model that organizes data description data semantics & consistency constraints of data.

Data emphasizes on what data needed & how it should be organized instead of what operations will be provided on data.

NoSQL databases types based on their data model

Key value pair based

Graph based

Column based

Document oriented

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① key value pair based

Data is stored in key/value pairs

Every data is stored in database as a key value pair consisting of attribute name (as a key) and value.

Example of key value
Redis - Diak

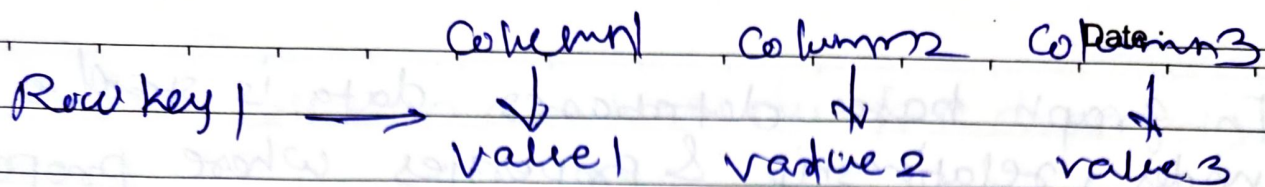
key	Name
Name	Richard
DoB	2-8-1988
Address	32-W NYC

② Column oriented database

- It works on column & is based on big table
Paper by google.
- Every column treated as separately
- The values of single column databases are stored continuously.

These are used to manage data warehouses
business intelligence, CRM library

Eg. HBase, Cassandra etc



Here ~~Row~~ row keys are used for grouping the data columns but actual data is stored in columns.

③ Graph based databases

A graph type database stores entities as well the relations amongst those entities. Relational databases whose table are loosely connected.

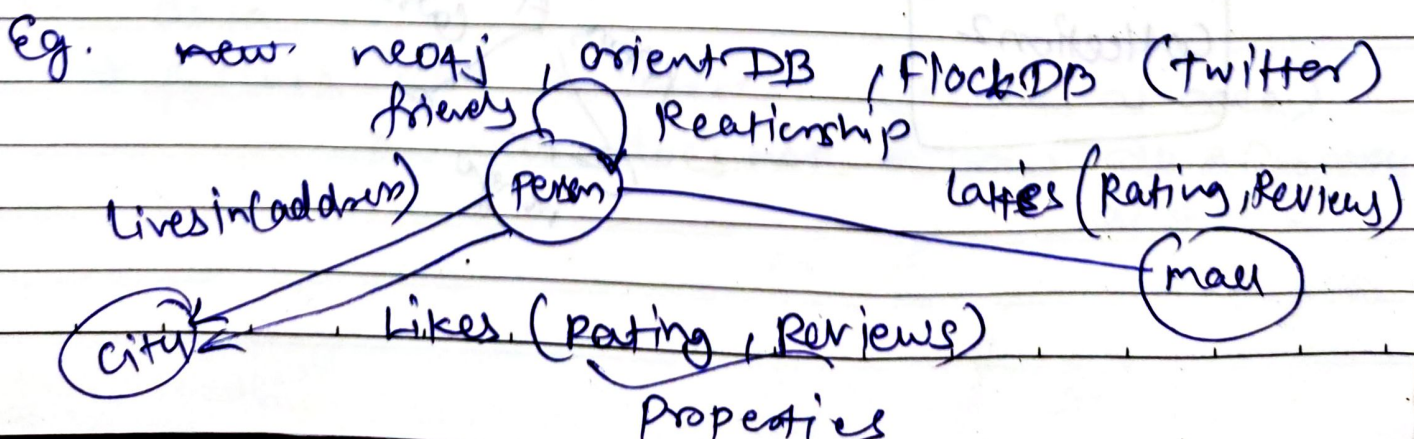
Graph database is multirelational in nature.

Entity stored as a node

Relationship as a edge & edge gives a relationship between nodes. Every node & edge have a unique identifier.

It is schema less

Uses → Social n/w and spatial data



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In graph base databases data is stored as nodes, relationship & properties where properties can be of nodes as well as of relationships.

④

Document oriented

Document oriented stores data, in JSON, BSON & XML documents

In this documents can be nested.

Particular elements can be indexed for faster querying

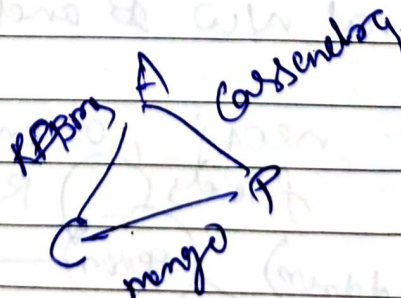
These databases are used dynamic schema & characteristics

Ex: MongoDB CouchDB (Couchbase)

Database

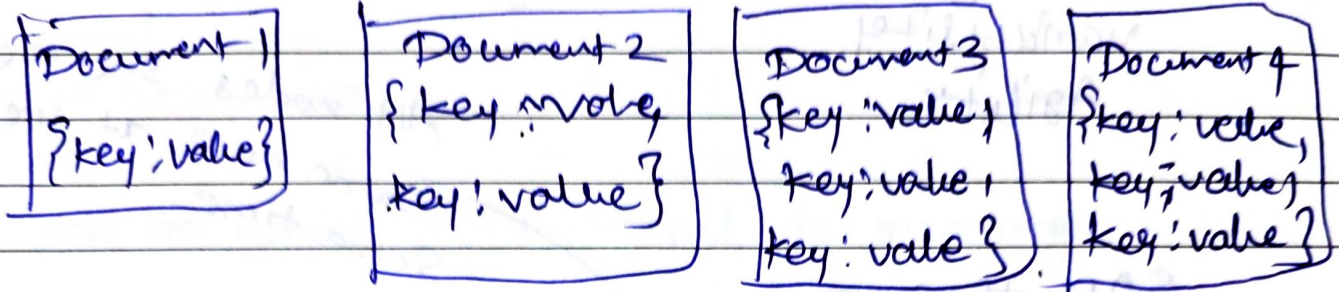
Collection1

Collection2



Each collection can have multiple documents

Collection



It is basically hierarchical version of key-value-based databases.

```
{
  ID: 200
  Product: mobile
  Amount: 10000
}
```

- Non relational database
- Doesn't have predefined schema
- Doesn't store data in tables
- Generally used to store big data & real time data
- follows CAP theorem (Any 2 at once)
- No SQL does not follow all 3 Property at a time

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No sql business drivers

volume

velocity

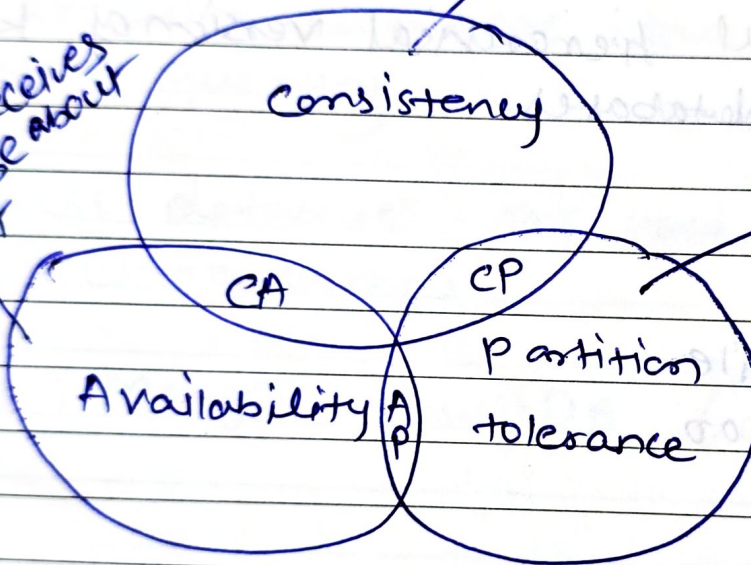
variability

Agility

All nodes see the
same data at the
same time

CAP theorem

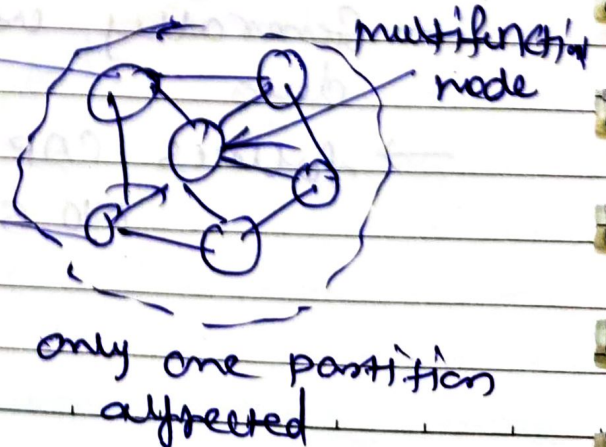
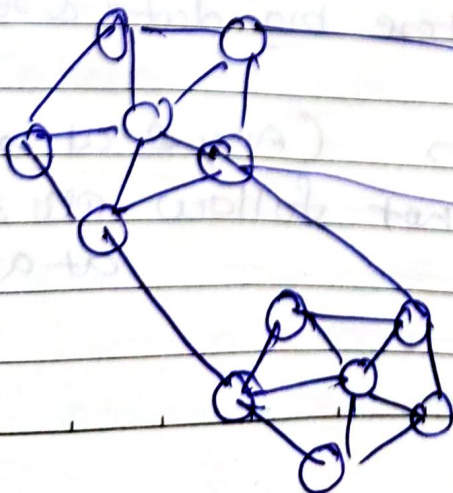
Every request receives
a response about
it
whether
succeeded
or failed



The system
continues to operate
even if there is a
partition between
two nodes.

It is a concept that a distributed db system
can have 2 of the 3.

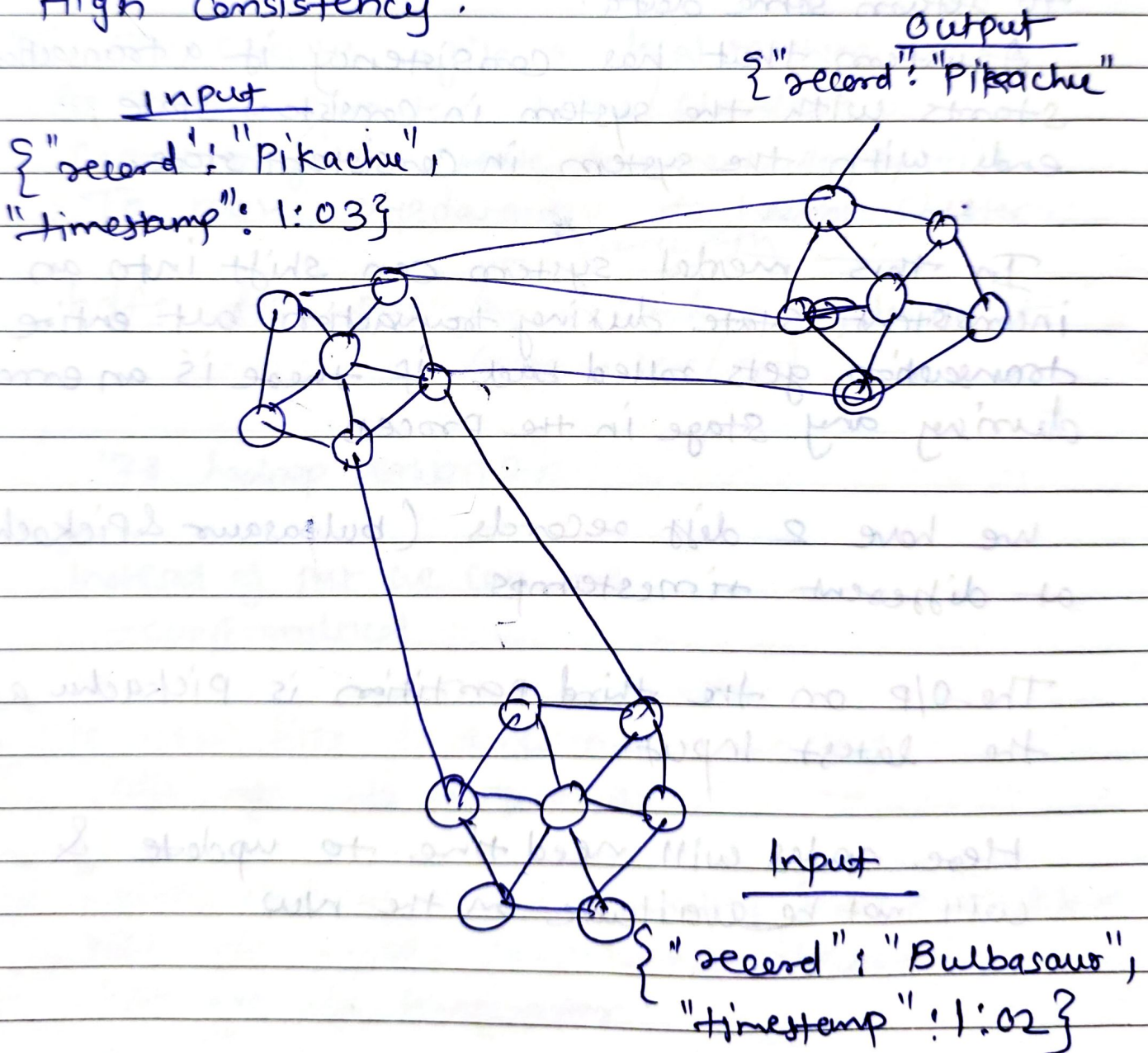
Partition tolerance



This condition states that system continues to run despite the no. of messages being delayed by the n/w between nodes. System that is partition tolerant can sustain any amount of n/w failure that doesn't result in a failure of entire n/w.

Data records are sufficiently replicated across combination of nodes & n/w to keep the system up through intermittent outages.

High Consistency.



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home/cloudera ↘

Create empdata.txt

This file contains my employee data like its name, address, designation etc.

The condition states that all nodes see the same data at the same time. Simply put performing read operations will return the value of the most recent write operation causing all the nodes to return same data.

A system that has consistency if a transaction starts with the system in consistent state, ends with the system in consistent state.

In this model system can shift into an inconsistent state during transaction but entire transaction gets rolled back if there is an error during any stage in the process.

We have 2 diff records (bulbasaur & Pickachu) at different timestamps

The O/P on the third partition is Pickachu as the latest input.

These nodes will need time to update & will not be available on the new

High availability

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This condition states that every request gets response on/ success/failure

Achieving availability in a distributed system requires that the system remain operational 100% of the time. Every client gets a response regardless of the state of any individual node in the system. Hence why, high availability isn't feasible when analyzing streaming data at high frequency.

CA	Cassandra, CouchDB, Riak, Voldemort
AP	Amazon DynamoDB, Google cloud Spanner
CP	MongoDB, Redis, Apache HBase.

There is need for company & organization to shift ~~relational data~~ ^{storing} traditional data to no sql unstructured data.

volume

velocity

variability

Agility

volume —

→ Need to scale up.

Extending Storage Capacity.

→ Need for faster processing

If data in large amount processing of that data resources ~~are~~ fulfilling the requirements.

→ Organizations shifted to from serial to parallel

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Entire data will be divided into cluster.

velocity

rate at which data is being generated.

→ Quick data generation

→ Random bursts in web traffic
~~very~~ low traffic

→ Slow response time.

→ Costly to deal.

To deal with response huge amount of resources are required.

variability (variety)

→ Diverse data type
data can be various dimension
audio 1d
audio & video 2d

→ Change in meaning or shape of data.

→ Compromise on availability, money & losing time.

Traditional database will not be able to tackle it.

Agility:

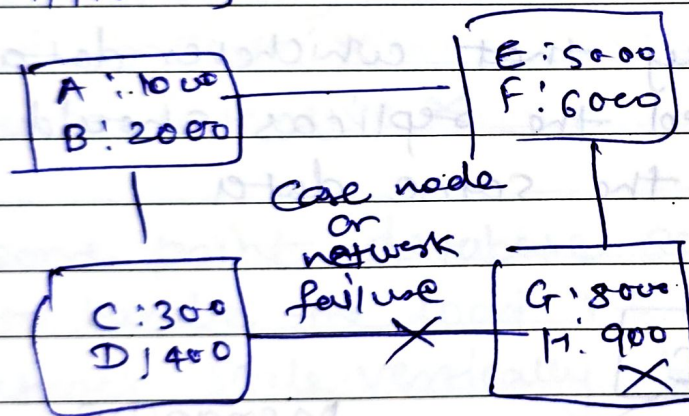
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- Quick & easy access & feeding of data.
- The time of retrieving & feeding of data is very high.

RDBMS contain many relations in a table

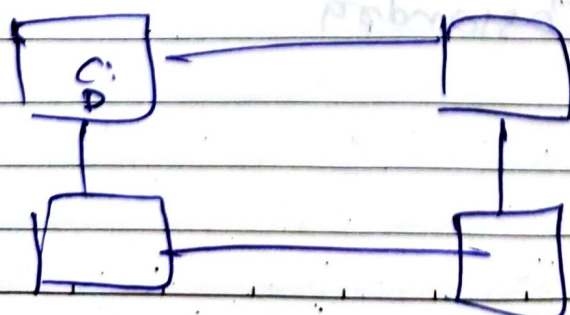
- RDBMS finds it difficult if data contains nests or many subgroups.
- which decreases response time.

Partitioning

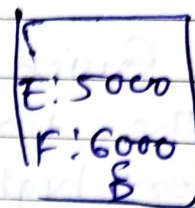
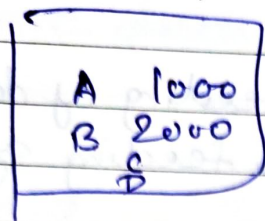


Availability

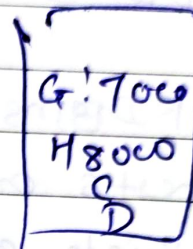
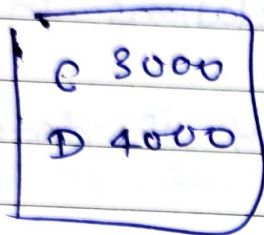
- All nodes must be operational
- The data must be available everywhere.
- Does not bother about whether the data is updated or not



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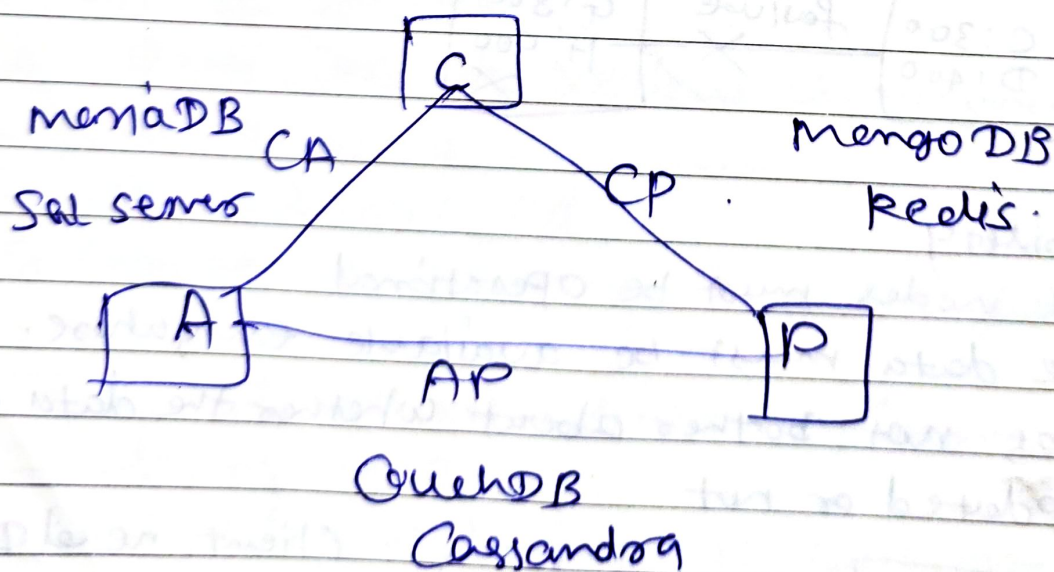


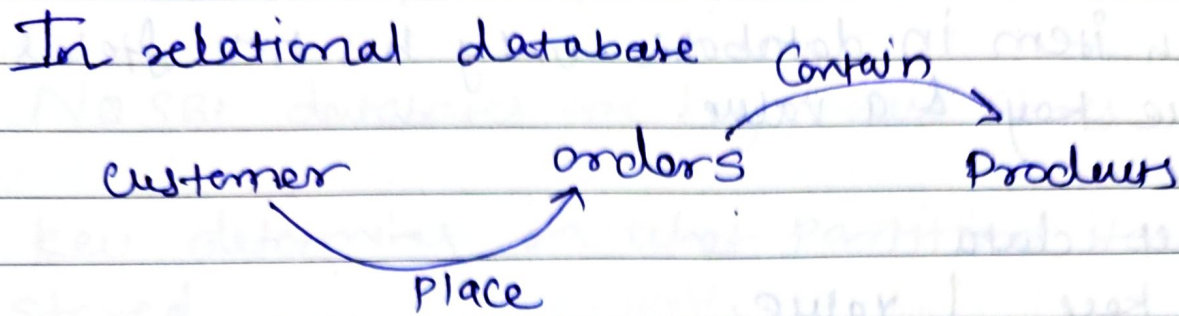
Client needs
data from c



Some other client has changed data
C from 3000 to 3500

Consistency says that whichever data has
got manipulated the replicas should be
updated with the same data





These 3 tables are tight logically, this tight organization is great for managing data
Cost of adms very high

Maintenance of relationship is very tedious process
require lot of memory & compute power

Data Consistency ✓

Hard to Scale X

Resource intensive X

At some point database server it won't be able to handle the load. we can scale relational databases scale vertically but not horizontally.

No SQL database can scale both vertically & horizontally.

vertical scaling is only possible to a certain extend, while horizontal scaling is much more powerful.

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NO SQL

Each item in database only has two fields
unique key & a value.

Product data

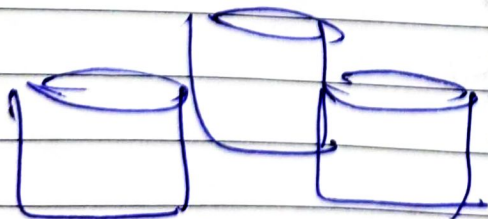
key	value
191213	{ name: "MacBook Pro", price: \$1000.21, description: "Laptop" }
191214	USB Microphone

If single database is not ~~eqs~~ enough to
store all your data or handle all the queries.
we can split the workload across two or more
servers.

Each server will then be responsible for only
a part of your database.

Apple runs a NO SQL databases that consist
of 75000 servers.

In NO SQL databases terms these part of your
database are called partitions & it brings up a



If your database is split across potentially
thousands of partitions how do you know where
an item stored?

That's where primary key comes in

NoSQL databases are key value stores

key determines on what partition item will be stored

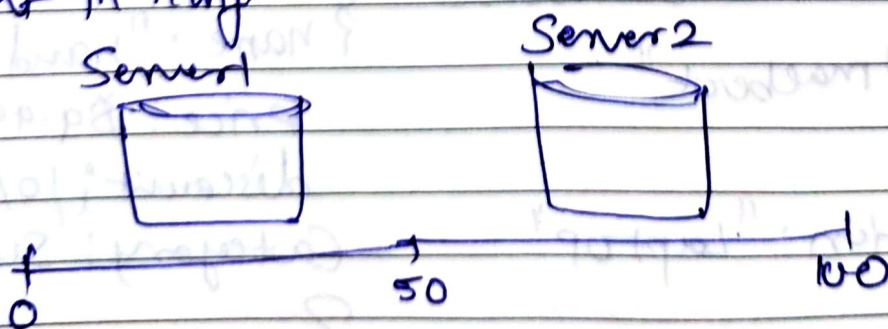
NoSQL database use hash function to convert each item's primary key into no fall between fixed range.
Say between 0 to 100

key	Hash	Value
19423245	23	MacBook Pro 13
42345691	87	USB microphone
360042312	23	Hand Sanitizer
8619308002	100	Detergent

This hash value & the range is then used to determine where to store an item.

If database requests one minimum then one server will handle load.

If that server is overloaded you can add a secondary server which means range will be split in half



Server 1 will be responsible for all items with a hash between 0 and 50, while Server 2 will store everything between 50 to 100.

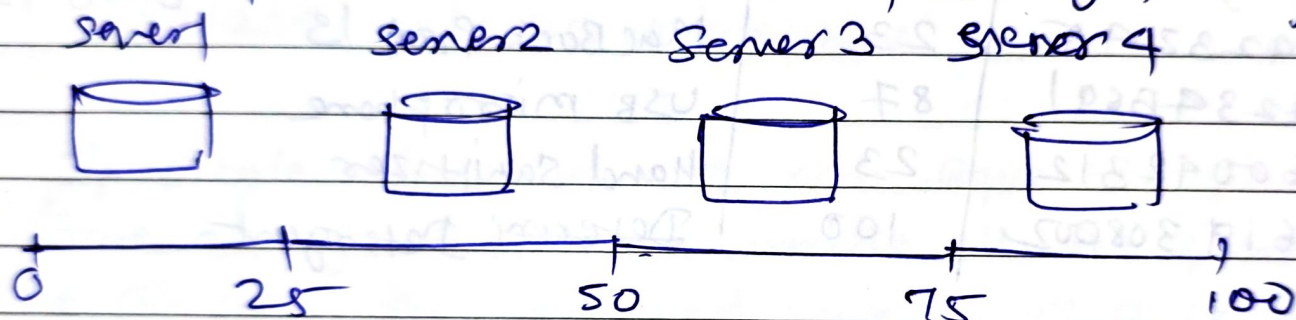
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This range is also called as keyspace.

It is a simple system that solves two problems

- ① where to store new items
- ② where to find existing one

All you have to do is calculate the hash of an item's key & keep track of which server is responsible for which part of the keyspace



Besides Scalability No SQL is Schemaless

Items in database don't need to have the same structure. Each one can be completely different

~~In a relational database~~

{ name: "macbook"

price:

description: "Laptop"

}

{ name: "hand sanitizers",

price: \$9.99;

discount: 10%.

Category: Survival"

}

~~Each one~~ In relational database you have to define your table's structure & then each item must conform to it.

Changing the structure could even lead loss of data

No SQL is allowing to retrieve items

Finding orders by id ✓ no problem

Get orders with amount $> \$100$ very inefficient

No SQL database stored data in partition, each partitioned is mirrored across multiple servers that way server can go down

When you write to new item to the database one of these mirrors will store new item.

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~~Mongo~~

Mongo is a command line shell
mongod is the Mongo Daemon its basically
host process for the database.

Comparison of MongoDB with MySQL

SQL Term Concepts

MongoDB Terms Concepts

database

database

tables

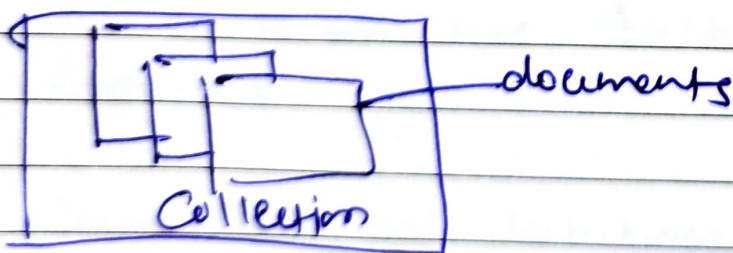
Collections

rows

document (BSON)

Columns

fields



```
{ name: "sam"  
  Age: 26  
  Status: "A"  
  groups: { "news", "reports" }  
}
```


What is Data Modelling

Why we use data modelling

Data models helps create a simplified.

Shared nothing architecture (SN) is a distributed computing architecture in which each update request is satisfied by a single node (processor / memory / Storage unit)

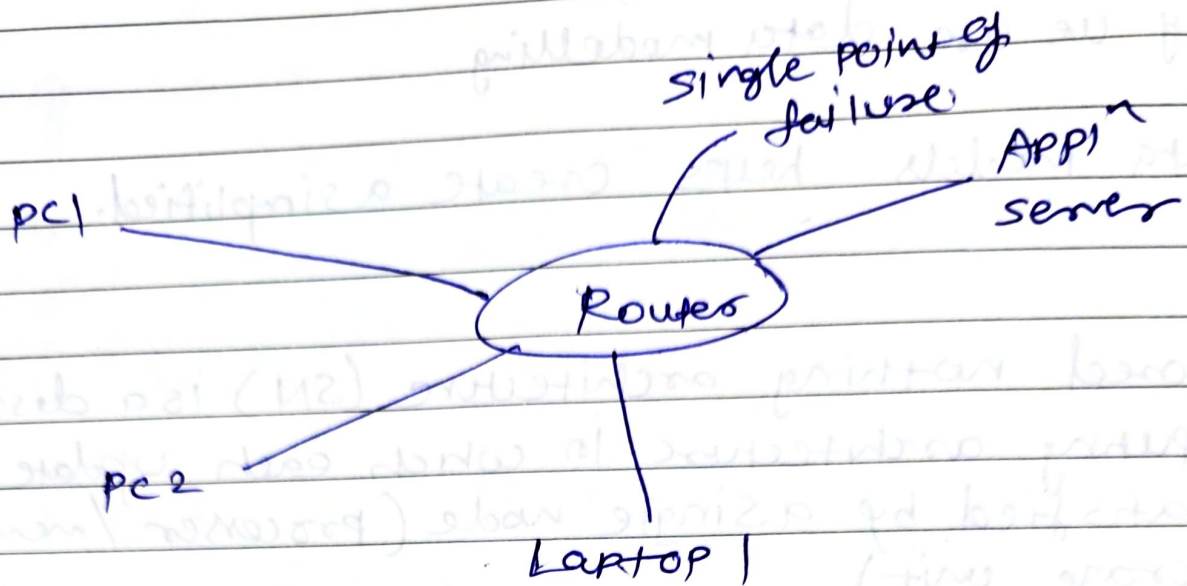
Nodes do not share (independently access) memory or storage.

one alternative architecture is shared everything in which requests are satisfied by arbitrary combinations of nodes

This may introduce contention, as multiple nodes may seek to update the same data at same time

SN eliminates single point of failure (SPOF) allowing the overall system to continue operating despite failures in individual nodes & allowing individual nodes to upgrade without a system wide shutdown.

SPOF is a part of a system that, if it fails will stop the entire system from working.



Shared nothing architecture is a distributed computing architecture in which each node is independent & self sufficient & there is no single point of contention across the system.

Advantage of SN versus a central entity that controls the n/w (a controller based architecture) include eliminating any single point of failure allowing self healing capabilities & providing an advantage with offering non disruptive upgrades.

Shared nothing is popular for web development because of its scalability.

A pure SN system can scale almost infinitely simply by adding nodes in the form of inexpensive computers since there is no single bottleneck to slow the system down.

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Share Everything

IP n/w

DB Server

Disk

eg. UNIX FS

Share Disks

IP n/w

DB DB DB DB

SAN DISKS

ORACLE

Share Nothing

DB DB DB DB

Local storage

Disk Disk Disk Disk

HDFS

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only specific big data manages
by NoSQL

Explain how NoSQL manages Big data

- NoSQL limits the support of join queries
- characterizes easy creation & high processing, speed, Scalability & storability of much higher magnitude of data
- It sacrifices the support of ACID Properties & instead supports CAP & BASE Properties.
- It provides data processing scales horizontally & vertically
- Big data solution needs Scalable storage of terabytes of data & petabytes of data
- Dropping of support for database joins.
- Storing data differently on several distributed servers (data nodes) together as a cluster
- A solution such as CouchDB, DynamoDB, MongoDB or Cassandra follows CAP theorem
- It makes transactions faster & easier to scale.
- The solution must also be Partitioning tolerant

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What are types of big data problems

- ① Big data needs scalable & distributed clusters
- ② No SQL database is open source & that is its greatest strength & its greatest weakness because then (not many standards define)
- ③ No stored procedures
- ④ GUI made tools & are not available in market
- ⑤ It ~~see see~~ sacrifices database ACID Properties.

Shared nothing architecture for big data tasks.

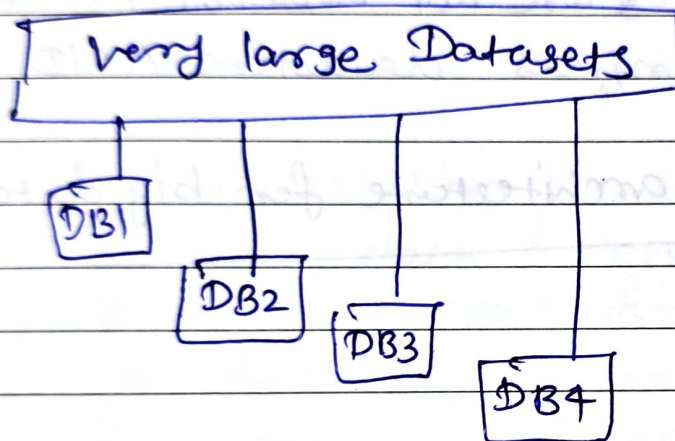
Shared memory

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Explain 4 distribution models in SN architecture

① Single server model
Simplest distribution option for NOSQL data store.

② Sharding very large Data bases Databases



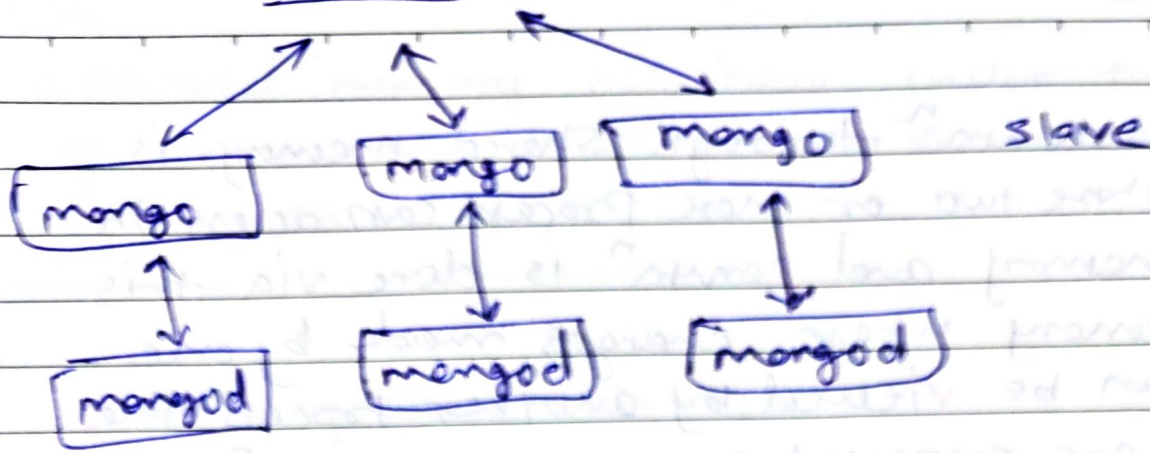
Different DB present in different server hence implementing sharding.

③ Master Slave distribution model.

- A node serves as a master or primary node.
- & other nodes are slave nodes.
- master directs the slaves
- When a process updates the master, it updates the slaves also.
- cluster based has greater complexity.

Client

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④ Peer-to-peer distribution model

- All replication nodes accept node request & send the responses
- All replicas function equally
- Node failure do not cause loss of write capability
- The data distributes among all the nodes in the cluster.

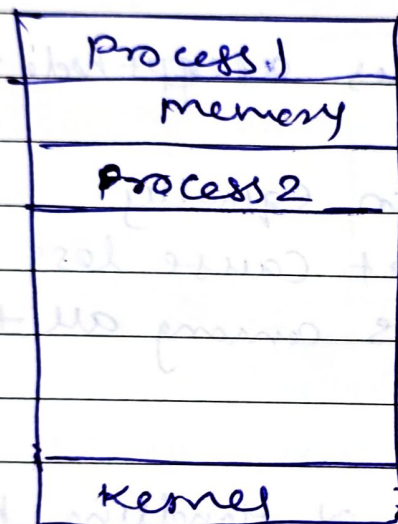
⑥ Mention some ways of Handling big data problems

- ① Evenly distribute the data on a cluster using the hash keys
- ② Use replication to horizontally distribute
- ③ Moving queries to the data not data to the queries
- ④ Queries distribution to multiple nodes

Sharding is a type of database Partitioning in which a large dataset is divided or partitioned into smaller data & diff nodes.

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Inter process commⁿ through shared memory is a concept where two or more process can access the common memory and commⁿ is done via this shared memory where changes made by one process can be viewed by another process. The processes are expected to exchange infoⁿ through the use of these shared variables.



- ① Create the shared memory segment or use an already created shared memory segment (`shmget()`)
- ② Attach the process to the already created shared memory segment (`shmat()`)
- ③ Detach the process from the already attached shared memory segment (`shmdt()`)
- ④ Control operations on the shared memory segment (`shmctl()`).